

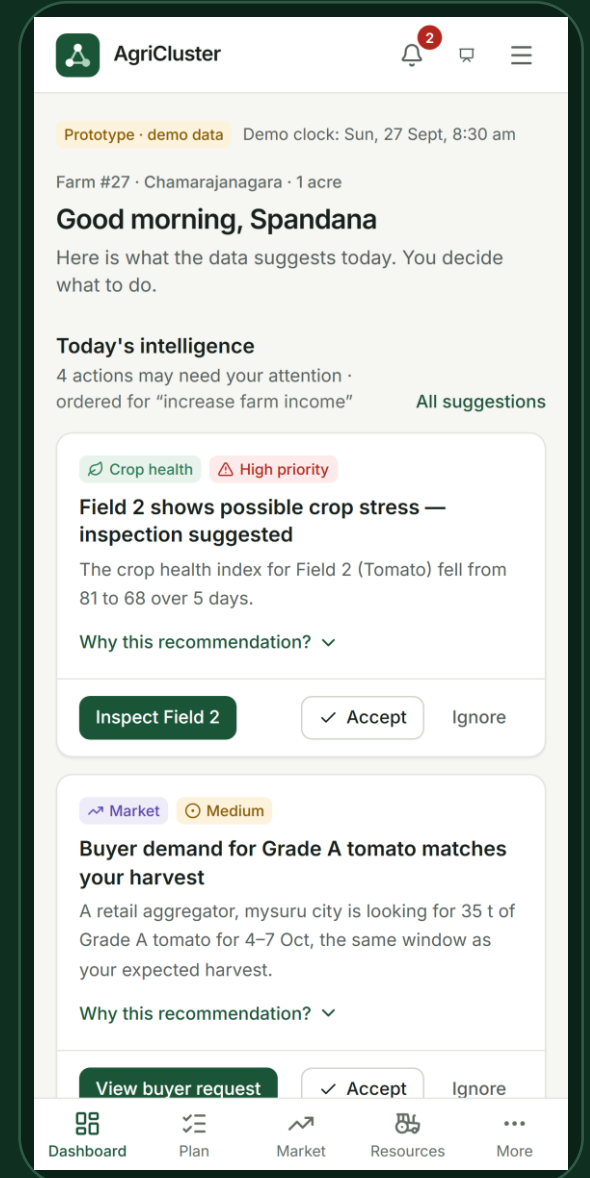


AgriCluster

Right Crop. Right Technology. Right Resource. Right Investment. Right Support.

Guides a small farmer from farm information to a buyer: the right crop, method, investment and support, with machinery, labour, experts and buyers shared across a cluster of nearby farms.

Competition prototype · Demonstration cluster: Mysuru Vegetable Cluster, Karnataka



Small farms face big decisions alone

Fragmented decisions

Each farmer decides alone when to irrigate, spray and harvest, often without timely data.

Fragmented resources

Tractors, labour and technology are hired or bought farm by farm, and everyone needs them in the same week.

Fragmented information

Weather, soil and crop signals rarely reach the field in a form a farmer can act on today.

Fragmented markets

Small lots mean weak bargaining power, uncertain buyers and avoidable post-harvest loss.

OUR DEMONSTRATION CLUSTER

128

farmers

312

acres, about 2.4 acres per farm

Tomato · Chilli · Onion · Leafy vegetables

Fictional demonstration data, Mysuru district

Smart-farming tools are built for one farm at a time

Weather apps

Solves Tell you it may rain.

Misses What that means for today's irrigation.

Farm IoT dashboards

Solves Detailed field data.

Misses A sensor kit per 2-acre farm is hard to justify.

Crop advisory apps

Solves General crop guidance.

Misses Your field's actual condition.

Equipment rental

Solves Finding a tractor.

Misses That 14 farms need one on the same day.

Market platforms

Solves Listing a crop.

Misses Pooling small lots into buyer-sized supply.

AI chatbots

Solves Answering questions.

Misses The farm data and follow-through behind a decision.

Each tool solves one piece. None coordinates the farms around it.

AgriCluster makes the cluster smart, not every farm

A shared intelligence and coordination layer for a group of neighbouring farms.

Guide

From farm details to the right crop, method, investment and cropping plan, with every suggestion explained.

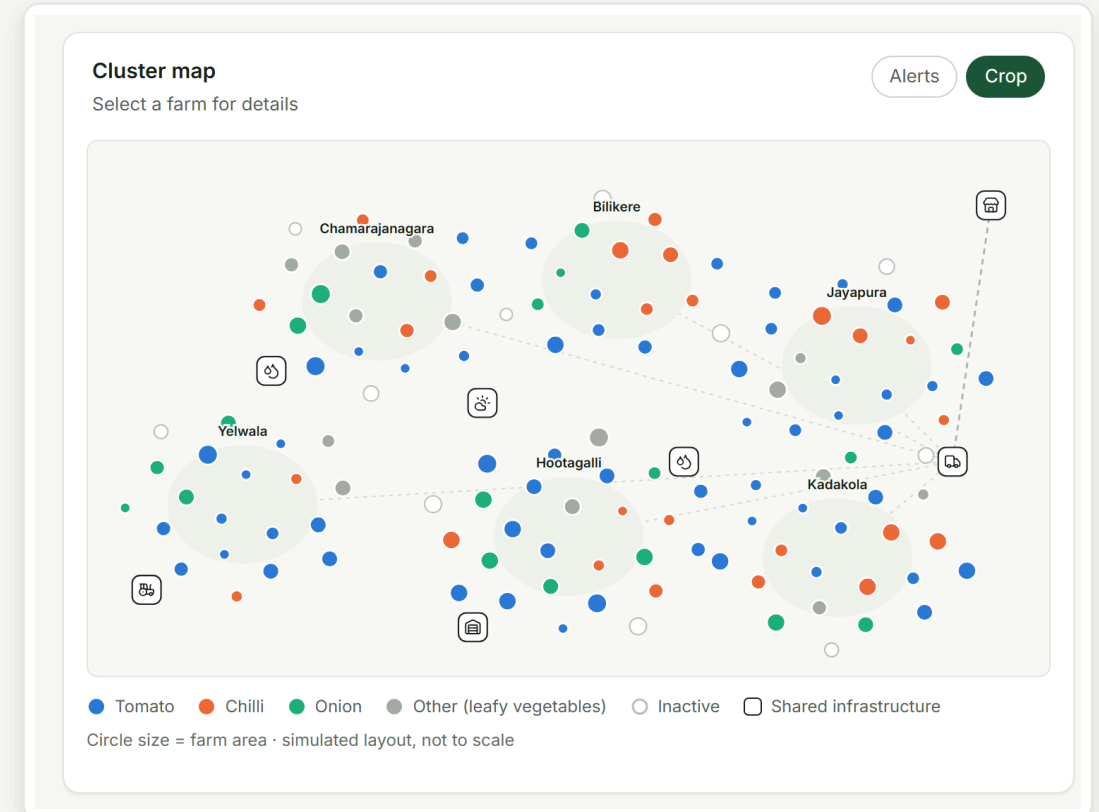
Connect

Farmers, machinery and technology providers, labour, experts, buyers, and government and private support, in one cluster.

Coordinate

Shared resources, pooled harvests matched to buyer demand, and a notification for every request and reply.

The platform recommends. The farmer decides.



128 farms in six villages, with shared water sources, machinery, cold storage and a collection centre

From farm information to a buyer, one decision at a time



SPANDANA'S SEASON IN THE PROTOTYPE

- Farm & goal**
1 acre near Chamarajanagara, red soil, drip, ₹1,50,000 to invest.
- Crop & method**
Tomato is the best match, with the reasons shown. Precision farming suits her drip and solar.
- Investment & plan**
₹1,50,000 split across ten costs; tractor and sprayer rented, not bought. First harvest around 4 October.
- Harvest & buyer**
2,000 kg of Grade A tomato listed. She chooses what buyers see; a buyer requests, she decides.

Good fit Best match

Choose tomato

Tomato

Kharif / late kharif (Jun–Oct)

Water Moderate	Growing period 98–112 days	Investment for your land ₹1,20,000–₹1,60,000	Labour High
Harvest Multiple pickings over 6–8 weeks	Technology fit High	Demand High	Indicative yield 8,000–14,000 kg

Retailers Restaurants Processors Wholesalers

Why are we suggesting this crop? ^

✓ Grows well in red soil like yours.

✓ Its water need (moderate) matches what you have.

✓ Your drip system suits it.

✓ Your ₹1,50,000 covers the lower end of the typical cost.

✓ You're growing tomato now, so you know it.

✓ Higher income potential per acre than most options (indicative).

✓ Buyers in the cluster are asking for it.

✓ Benefits from the technology you're comfortable with.

⚠ Price swings between seasons

Crop step: indicative ranges and the reasons behind the suggestion, never a guaranteed income

One signal on one farm becomes a pattern across the cluster

DATA COMBINED

Farm & crop stage

Soil moisture

Weather

Water & energy

Machinery

Labour

Experts

Market demand

ACROSS THE CLUSTER TODAY

24 farms plan to irrigate before tonight's rain

6 tomato farms show possible crop stress

14 tractor requests for 9 tractors tomorrow

42 t of tomato expected against 35 t of buyer demand

← Intelligence

Water

High priority

Field 1

Consider delaying Field 1 irrigation by 24 hours

WHAT IS HAPPENING?
Rain is expected in about 10 hours and Field 1 soil moisture is already adequate.

WHY DOES IT MATTER?
Irrigating just before 18–25 mm of rain uses water and pumping energy the crop is unlikely to need, and can waterlog the root zone during fruiting.

DATA BEHIND THIS SUGGESTION

Rain forecast	82% chance, 18–25 mm from 6:30 pm	Demo data
Soil moisture	64% (crop needs ≥ 55%)	Simulated sensor
Crop stage	Tomato · Fruit development	Demo data
Irrigation scheduled	Today, 5:00 pm	Demo data

POTENTIAL IMPACT
Could avoid about 15,000 L of water and ~4 kWh of pumping (indicative estimate).

How smart irrigation advice works

- 1 Soil moisture in your field is monitored.
- 2 The weather forecast is checked for rain.
- 3 Your crop and its growth stage set how much water it needs.
- 4 AgriCluster compares these against your irrigation schedule.
- 5 A suggestion is generated, with the reasons shown.
- 6 You decide whether to act.

Every suggestion shows what is happening, why it matters, and the data behind it, with each value's source labelled.

Demonstration data

Seven roles, one connected cluster

Farmer

Plans the season, finds support, requests resources, lists harvests.

Buyer

Searches crops, sees farmer-approved details, requests farmers.

Machinery & tech owner

Lists equipment and services; accepts bookings.

Labour

Lists skills, crops, rates and transport; accepts work.

Expert

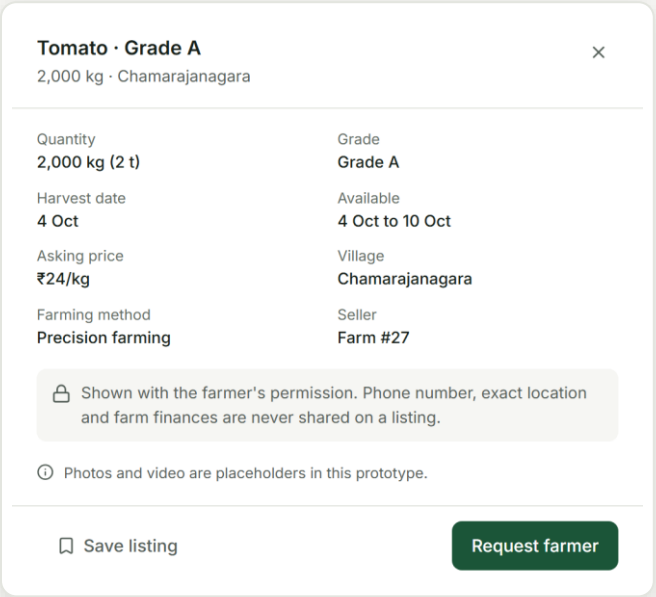
Answers questions and books consultations, across 11 fields.

Community

Runs farmer groups, events and success stories.

Cluster manager

Sees all 128 farms; coordinates water, resources, supply.



What a buyer sees on "View crop": farmer-approved details only

Privacy by role

Buyers see only what the farmer approved on each listing: never her phone number, finances or exact location. Labour sees the job's crop and village. Experts see the question and the field it is about.

Connected by notifications

Every request, offer and reply notifies the other side, from "Buyer requested your tomato crop" to "Expert consultation confirmed".

Impact we will measure against a baseline

15–20%

less avoidable water use

Pilot target

10–15%

better input efficiency

Pilot target

10%

less crop loss

Pilot target

5–10%

better market realisation

Pilot target

Pilot targets are illustrative goals, not guaranteed outcomes. Improvements are claimed only against a measured baseline season.

WHAT THE PROTOTYPE ALREADY SHOWS

≈15,000 L

water and ≈4 kWh pumping avoided on 1 acre by one accepted irrigation suggestion (indicative)

42 t vs 35 t

pooled tomato supply against buyer demand, visible a week before harvest

14 vs 9

tractor requests against available tractors, flagged a day ahead

Revenue follows coordination

REVENUE STREAM	WHO PAYS	MODEL
Cluster services	FPOs, cluster organisations, agriculture programmes	Subscription per cluster
Equipment marketplace	Equipment owners and renters	Commission on completed bookings
Expert services	Farmers or sponsoring institutions	Commission or subscription
Market linkage	Buyers	Service fee on agreed deals
Technology & analytics	Institutions and agricultural organisations	Implementation fees and cluster analytics

Start simple. The pilot runs on a cluster subscription and a marketplace commission. The rest follows as volume grows.

Pilot: one cluster, measured over a season

100+

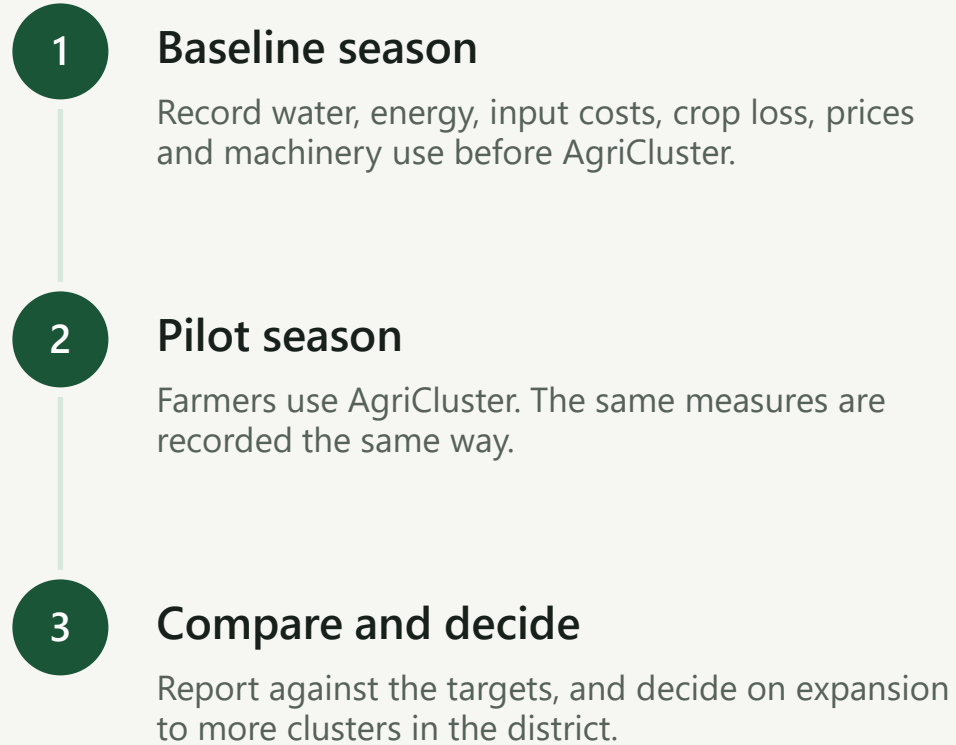
farmers

250+

acres

1

cluster



WHAT WE TRACK

- Water use
- Energy use
- Input costs
- Crop loss
- Machinery utilisation
- Market realisation
- Farmer adoption

The intelligence layer scales. The farms stay small.



Real database, simulated sensors

Runs today on a cloud database (MongoDB Atlas) that keeps each space private. Sensor readings are simulated; real sensors plug into the same interfaces later.

Same engine, every cluster

A new cluster reuses the intelligence rules, roles and screens. Only local data changes.

Network effects

Each new cluster brings more buyers, equipment providers and experts to every member.

Connected agricultural clusters across India

Today

Fragmented farms
Fragmented information
Fragmented resources
Fragmented decisions
Fragmented markets



With AgriCluster

Connected farms
Shared resources
Integrated data
Cluster intelligence
Coordinated decisions and markets



Future

More efficient, more resilient
agricultural clusters, where farmers
keep the decisions and technology
does the coordination.



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The platform recommends. The farmer decides.